

Syllabus

Spring semester 2026

Course Title	General Chemistry 2		Course Code		CB3103		Credit	4	Classroom	3303
			Type		Elective (AP)		Hours	4		
Teacher	Name	Class no.	Office	Office Hour*		Contact		E-mail		
	Park, Seonhwa	1	5604	M3, Tu3, F3		378		spark1122@ksa.kaist.ac.kr		

*Office Hour: Time for students to ask questions and discuss lecture or lab topics with teachers for deeper understanding.

1. Course Description

Based on what is learned in the Chemistry and Experimental course, a first-year required course at KSA that corresponds to the high school chemistry curriculum, students learn chemistry contents equivalent to the general chemistry level of a university. Students who have completed General Chemistry 1 and General Chemistry Experiment 1 as prerequisite subjects can take this course. The contents to be learned in this course are as follows: (1) Thermodynamic Processes and Thermochemistry, (2) Spontaneous Processes and Thermodynamic Equilibrium, (3) Chemical Equilibrium, (4) Acid-Base Equilibria, (5) Solubility and Precipitation Equilibria, (6) Electrochemistry, (7) Chemical Kinetic, (8) Nuclear Chemistry, (9) Structure and Bonding in Solids, (10) New materials - Inorganic Materials, Polymers, and Soft Condensed Matter.

2. Core Target Competencies

Subject Core Competencies						
Subject comprehension	Scientific thinking	Problem solving ability	Communication skills	Ability to explore	Teamwork	Creativity
Through team problem-solving activities, students will develop an understanding of the basic concepts of chemical equilibrium and reaction kinetics of various chemical reactions and the ability to solve related problems. During the problem-solving activity process, students collaborate with team members to solve problems, develop communication skills through discussion, and cultivate the ability to think logically using scientific basis. Additionally, through literature review, we understand trends in the development of new inorganic materials, polymers, and soft condensed materials in the chemical field.						

3. Textbooks and References

- Textbook: Principles of Modern Chemistry (8th Ed.), by Oxtoby, Gillis and Butler (Cengage Learning, Inc.)
- References: Internet resources and academic papers

4. Class Method

Students form teams and conduct classes through team problem-solving activities. Students think about learning objectives through a discussion process and learn core scientific concepts and scientific knowledge while solving given problems through collaboration.

- Autonomously form small groups and discuss learning objectives.

B. Collaborate in small group activities to solve problems together and present your understanding to your peers.

C. Share the class materials you solved together and write reflective essays based on what you understood.

5. Evaluation

A. Evaluation Criteria

Problem-Solving Activity	Presentation on New Materials	Midterm Exam	Final Exam	Attendance and Attitude	Total
10	5	40	35	10	100

B. Evaluation Method

(1) Written Exams (75%)

- Comprehension of course contents are assessed through midterm exam (40%) and final exam (35%).
- The exam questions are written in English and are mainly descriptive based on the content learned.

(2) Team problem-solving activities (10%)

- Evaluate activities that involve collaborating with team members to solve problems and present them.
- Evaluate problem-solving activity sheets (submission deadlines and content fidelity)
- Evaluate reflective essays by unit topics (submission deadlines and content fidelity)
- Failure to submit the problem-solving activity sheet by the deadline will result in a 0.2 point deduction per activity sheet.

(3) Presentation on trends in new materials research (5%)

- Each student researches new materials and gives a 10-minute presentation.
- Submit a report based on your own research on new materials.

(4) Attendance and Attitude (10%)

- If you engage in behaviors that interfere with learning, such as chatting, napping, or studying other subjects, 0.5 points will be deducted from your learning attitude.
- Students who receive a deduction in learning attitude will receive a 0.5 point discount for learning attitude if they contribute positively to team learning and show enthusiasm during class.
- Being late will be penalized -0.5 point per each class.
- 1 point deducted for each unauthorized absence.

6. Criteria for Awarding Credits

- ☞ Final letter grade will be awarded based on an absolute evaluation of the AP grading criteria and the total number of points received by the student.
- ☞ Students who are absent three or more times without a valid reason will receive an F grade.
- ☞ If you are uncooperative in presentations and activities, you may receive an F grade depending on

the circumstances.

☞ **Students with excellent academic achievement are recommended for the Academic Excellence Award and the Principal's Award, and are recorded according to core competencies in the Subject Details. However, students with poor academic attitude or poor attendance are excluded.**

- Attendance for at least three-quarters of the total class hours is required to earn credit.
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- Calculation of the basis score for awarding credit to absentees : For officially excused absences or medically excused absences, the recognized score for a missed exam will be calculated as follows: (Percentile rank equivalent to the total score obtained in other exam of the same course taken by the student) × (100% for officially excused absences; 80% for medically excused absences).
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- If a basis score cannot be calculated, the grading method : If a basis score cannot be calculated, grading will follow the Academic Performance Management Regulations.
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- ※ Refer to the "Academic Performance Management Regulations" for the percentage of credit points applied.

7. Weekly Plans

Week (period)	Topic and Contents	(Notes)
Week 1 2.23.~3.1.	Class Orientation Chapter 12. Thermodynamic Processes and Thermochemistry	
Core Competencies	♦ Subject comprehension, Scientific thinking, Problem solving ability, Communication skills, Teamwork	
Week 2 3.2.~3.8.	Chapter 12. Thermodynamic Processes and Thermochemistry Chapter 13. Spontaneous Processes and Thermodynamic Equilibrium	3.2. Substitute Holiday
Core Competencies	♦ Subject comprehension, Scientific thinking, Problem solving ability, Communication skills, Teamwork	
Week 3 3.9.~3.15.	Chapter 13. Spontaneous Processes and Thermodynamic Equilibrium	
Core Competencies	♦ Subject comprehension, Scientific thinking, Problem solving ability, Communication skills, Teamwork	
Week 4 3.16.~3.22.	Chapter 14. Chemical Equilibrium	
Core Competencies	♦ Subject comprehension, Scientific thinking, Problem solving ability, Communication skills, Teamwork	
Week 5 3.23.~3.29.	Chapter 14. Chemical Equilibrium Chapter 15. Acid-base Equilibria	
Core Competencies	♦ Subject comprehension, Scientific thinking, Problem solving ability, Communication skills, Teamwork	
Week 6 3.30.~4.5.	Chapter 15. Acid-base Equilibria	
Core Competencies	♦ Subject comprehension, Scientific thinking, Problem solving ability, Communication skills, Teamwork	
Week 7 4.6.~4.12.	Chapter 16. Solubility and Precipitation Equilibria	
Core Competencies	♦ Subject comprehension, Scientific thinking, Problem solving ability, Communication skills, Teamwork	
Week 8 4.13.~4.19.	Midterm Exams (4.14.~4.17.)	4.13. Regular Classes

Week 9 4.20.~4.26.	Chapter 17. Electrochemistry	
Core Competencies	◆ Subject comprehension, Scientific thinking, Problem solving ability, Communication skills, Teamwork	
Week 10 4.27.~5.3.	Chapter 17. Electrochemistry Chapter 18. Chemical Kinetics	
Core Competencies	◆ Subject comprehension, Scientific thinking, Problem solving ability, Communication skills, Teamwork	
Week 11 5.4.~5.10.	Chapter 18. Chemical Kinetics	5.5. Children's Day
Core Competencies	◆ Subject comprehension, Scientific thinking, Problem solving ability, Communication skills, Teamwork	
Week 12 5.11.~5.17.	Chapter 19. Nuclear Chemistry	
Core Competencies	◆ Subject comprehension, Scientific thinking, Problem solving ability, Communication skills, Teamwork	
Week 13 5.18.~5.24.	Chapter 21. Structure and Bonding in Solids	
Core Competencies	◆ Subject comprehension, Scientific thinking, Problem solving ability, Communication skills, Teamwork	
Week 14 5.25.~5.31.	Presentation on new inorganic materials, polymers, and soft condensed materials	5.25. Substitute Holiday
Core Competencies	◆ Subject comprehension, Scientific thinking, Problem solving ability, Communication skills, Teamwork	
Week 15 6.1.~6.7.	Presentation on new inorganic materials, polymers, and soft condensed materials	6.3. Local Election Day
Core Competencies	◆ Subject comprehension, Scientific thinking, Problem solving ability, Communication skills, Teamwork	
Week 16 6.8.~6.14.	Make-up class week	
Week 17 6.15.~6.19.	Final Exams (6.18.~6.19.)	

※ The above plans may change according to the academic schedule.